

Deep Dive: 19. Advanced: Generators

Concept Summary

Generators yield one item at a time instead of returning a massive list all at once. This saves massive amounts of memory.

Coding Structure

```
def gen():  
    yield value
```

Example Code

```
def counter():  
    yield 1  
    yield 2  
  
c = counter()  
print(next(c))  
print(next(c))
```

Code Explanation

- Line 2-3: The 'yield' keyword is like 'return', but it PAUSES the function instead of killing it.
- Line 5: We create the generator object.
- Line 6: We call next() to unpause it. It runs until it hits the first yield, spits out 1, and pauses.
- Line 7: We call next() again. It resumes from where it left off, spits out 2, and pauses.

Practice Problems

1. Write a generator function that yields infinite even numbers (using a while True loop).
2. Use a for-loop to iterate over your generator (make sure to break out of it before it crashes your PC!).